Signals and Systems
SET 214
Lecture (01)- Course Introduction

Dr. Ahmed Reda Mohamed[1]

[1]The MoE Key Laboratory of Artificial Intelligence & Bio-Circuits and Systems Laboratory (*BiCASL*), Department of Micro-Nano Electronics, Shanghai Jiao Tong University, Shanghai, China.

Outlines

- ❑ Course Information
- ❑ Course Assessment
- ❑ Course Content
- ❑ Course Long-Term-Vision
- ❑ IC Industry In Egypt
- ❑ References

Outlines

- ❑ Course Information
- ❑ Course Assessment
- ❑ Course Content
- ❑ Course Long-Term-Vision
- ❑ IC Industry In Egypt
- ❑ References

Course Information

- **Lecture's Time:**
  - 08.30: 10.30@ Tuesday
- **Lecture's Place:**
  - A308

- **Lectures Materials, Quiz, Assignment, and Project:**
  - Please, check ECU-LMS
  - Please, check Microsoft team

- **Contacts:**
  - Email : A.reda@ecu.edu.eg , Ahmed.reda@sjtu.edu.cn 
  - Whatsapp: 01020202427 

Outlines

- Course Information
- Course Assessment
- Course Content
- Course Long-Term-Vision
- IC Industry In Egypt
- References

Course Assessment

- **Final Exam [40%]**
  - You must get > **30%** of the paper grade or you will **fail** in the course regardless of your course work.
- **Lab, Quiz, and Assignment [30%]**
  - You must attend all course sessions otherwise you will **lose** some degrees
- **Midterm [20%]**
  - 1st midterm [**20%**]→7th week
- **Project [10%]**
  - **5** students



Forbidden

Copy or cheat during **Exams** and even doing the **Quiz** and **Assignment**

Outlines

- ❑ Course Information
- ❑ Course Assessment
- ❑ Course Content
- ❑ Course Long-Term-Vision
- ❑ IC Industry In Egypt
- ❑ References

Course Content (1/2)

❑ Course Objective

- ❑ Overview of important related topics on digital design systems to expand students' horizons about the digital design of integrated circuit or discrete element.

  The course consists of the ***six*** modules.

❑ Module 1: Signals and Systems

- ❑ Introduction. Signals and Classification of Signals: Continuous-Time and Discrete-Time Signals , Real and Complex Signals, Periodic and Nonperiodic Signals, Energy and Power Signals. Basic Continuous-Time Signals: The Unit Step Function, The Unit Step Function, Complex Exponential Signals, Sinusoidal Signals. Systems and Classification of Systems: System Representation, Systems with Memory and without Memory, Causal and Noncausal Systems, Linear Systems and Nonlinear Systems, Time-Invariant and Time-Varying Systems, Linear Time-Invariant Systems. Exercise.

❑ Module 2: Linear Time-Invariant Systems

- ❑ Introduction. Response of a Continuous-Time LTI System and the Convolution Integral: Impulse Response, Response to an Arbitrary Input, Convolution Integral, Properties of the Convolution Integral, Convolution Integral Operation. Exercise.

Course Content (2/2)

Module 4: Fourier Analysis of Continuous-Time Signals and Systems

- Introduction. Fourier Series Representation of Periodic Signals. The Fourier Transform. Properties of the Continuous-Time Fourier Transform: Linearity Time Shifting Frequency Shifting Time Scaling Time Reversal Duality Convolution Multiplication. The Frequency Response of Continuous-Time LTI Systems: Frequency Response, filtering, bandwidth.

Module 5: Laplace Transform and Continuous-Time LTI Systems

- Introduction. The Laplace Transform. The Region of Convergence. Laplace Transforms of Some Common Signals. Properties of the Laplace Transform. The Inverse Laplace Transform. The System Function and it prosperities.

Module 6: Sampling and Discreate-time signal

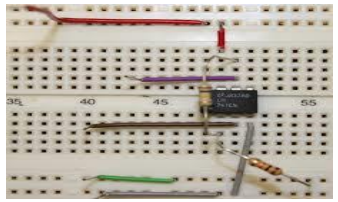
- Sampling Theory. Discreate-time signal classification. Basic discreate-time Signals. Discreate-time system types. Discreate-time signal LTI System and the Convolution Integral.

Outlines

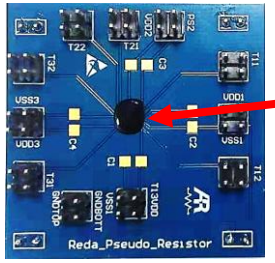
- ❑ Course Information
- ❑ Course Assessment
- ❑ Course Content
- ❑ Course Long-Term-Vision
- ❑ IC Industry In Egypt
- ❑ References

Course Long-Term-Vision (1/4)

Discrete Element



Bread Board

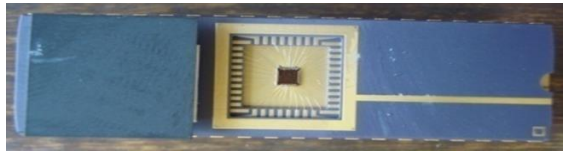


Chip-On-Board (COB)



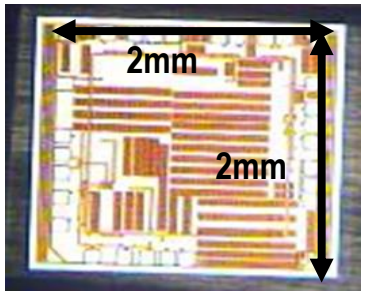
Printed Circuit Board (PCB)

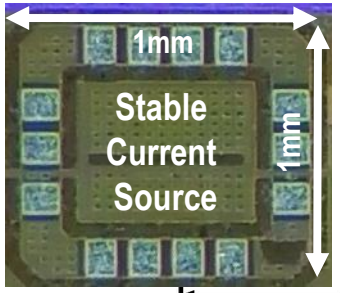
- ❑ Through-hole
- ❑ Surface mount

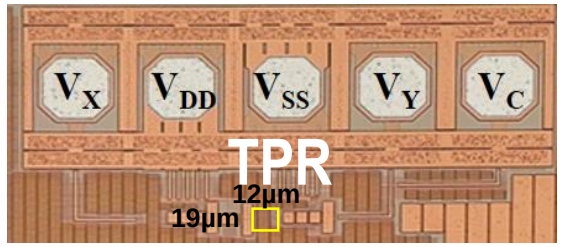


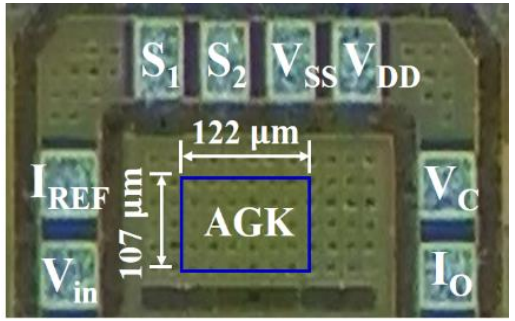
MOSIS (Granted)

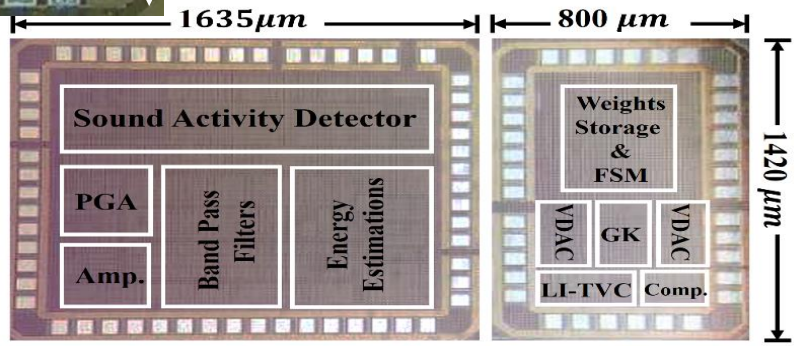
Vey large Scale Integrated Circuit







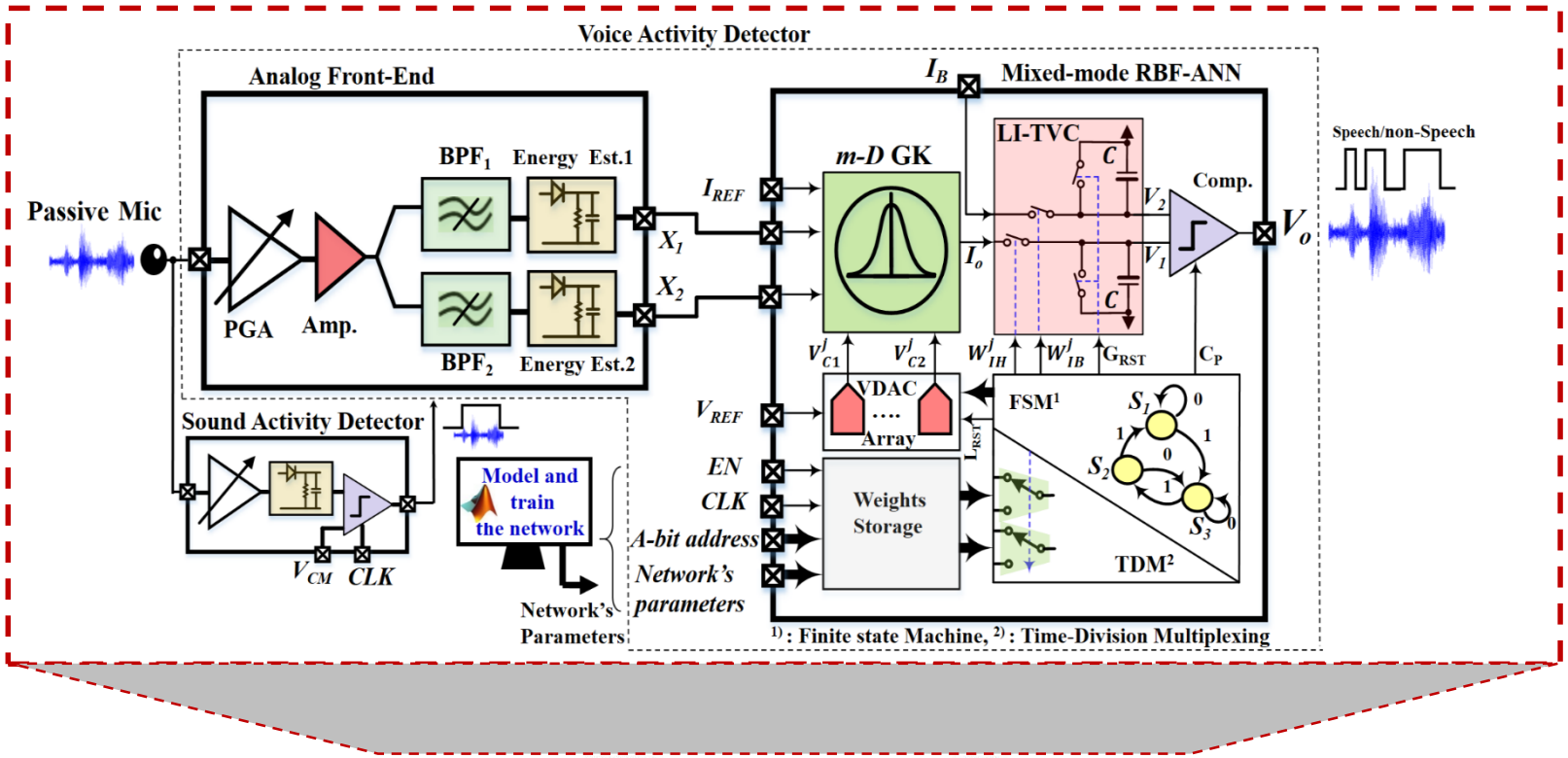


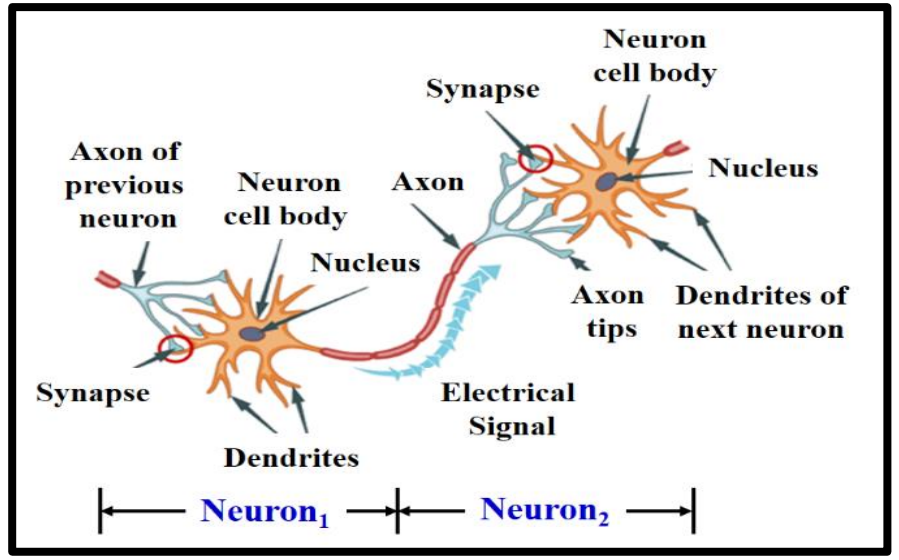


A. Reda

Item	Discrete Element	Very large scale Integrated Circuit
Area	Large	Small
Configuration	Easy	Complex
Price	Cheap	Expensive
Applications	Small production	Mass production
Power consumption	High	Low

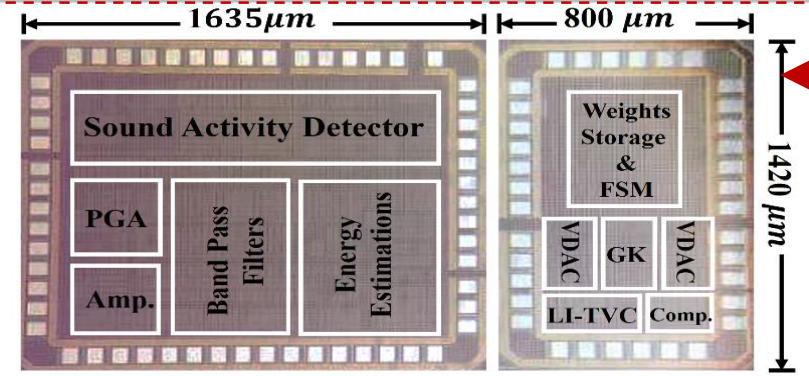
Course Long-Term-Vision (2/4)



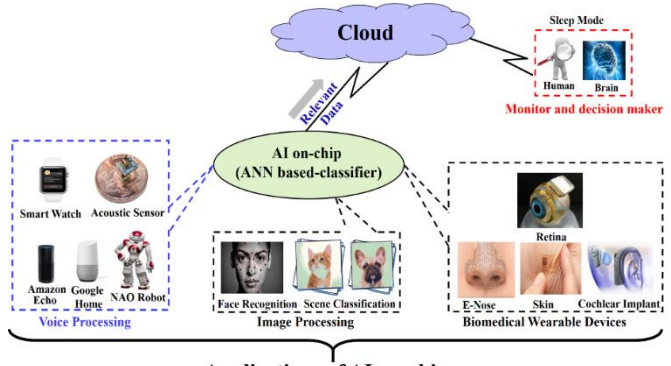


Brain biological system.



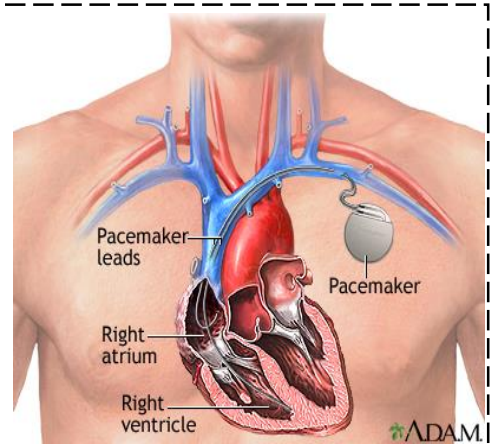


Artificial Neural Network On-Chip

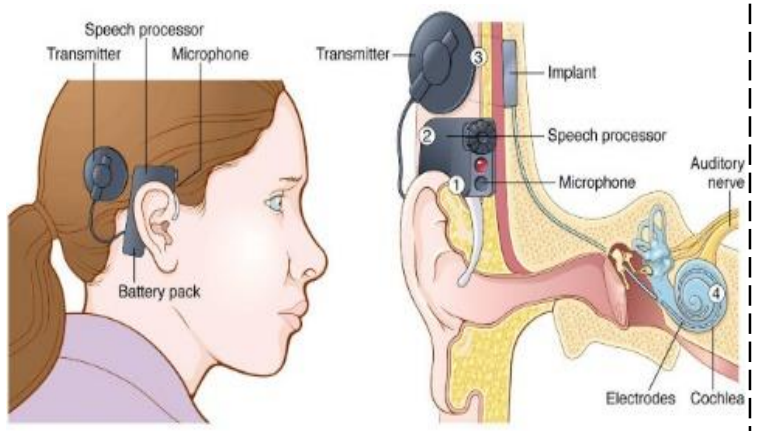


Course Long-Term-Vision (3/4)

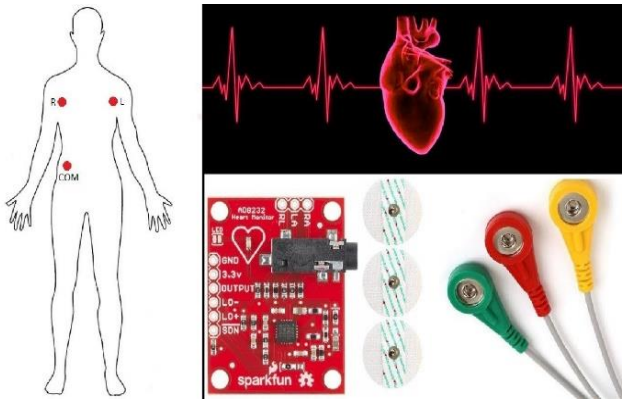
Electronics in Bio-medical Devices



Pacemaker



Cochlear implant

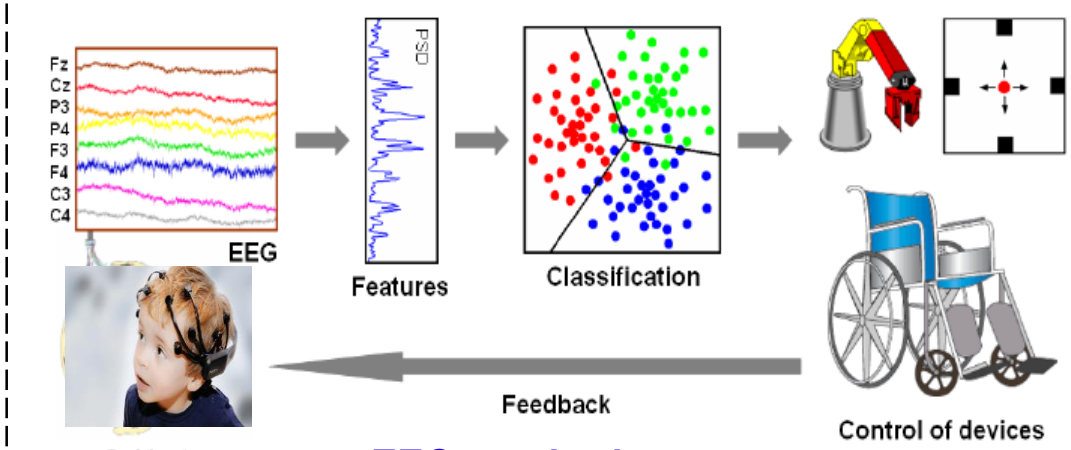


ECG monitoring

ECG	ElectroCardio-Gram
PPG	Photo-Plethy-smo Graphy
EEG	Electro-encephalo-gram



PPG monitoring

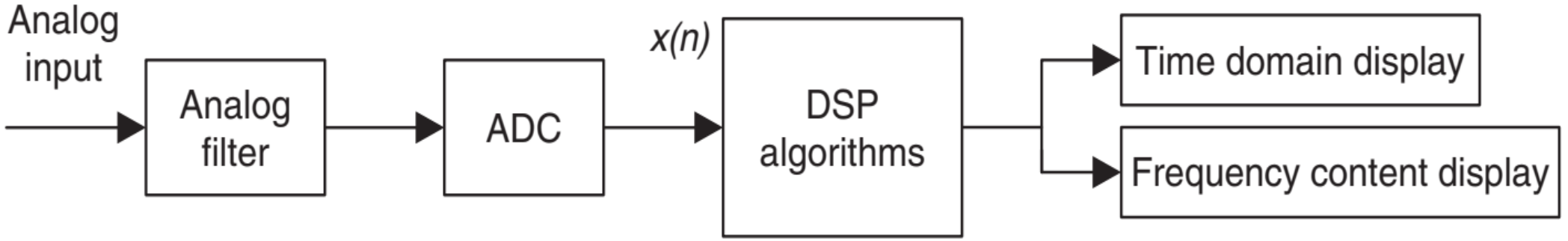


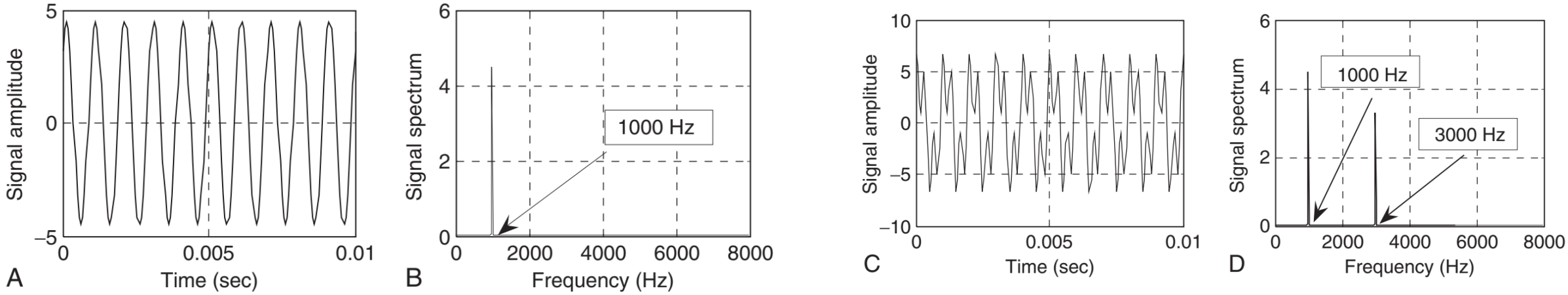
EEG monitoring



Course Long-Term-Vision (4/4)

Audio-Noise Cancellation





Audio signals and their spectrums

Outlines

- ❑ Course Information
- ❑ Course Assessment
- ❑ Course Content
- ❑ Course Long-Term-Vision
- ❑ Time_Plan and Project
- ❑ References

Time_Plan and Project (1/2)



1st week
Happy new academic year
Curriculum introduction

2nd week

3rd week
- Submit team members
- Submit Project name

4th week
Quiz1

5th week

6th week
Quiz2

7th week
Midterm

8th week
- Follow up Project

9th week

10th week
Quiz3

11th week

12th week
Quiz4

13th week
- Discussion
- PPt.,
- Demo and Poster

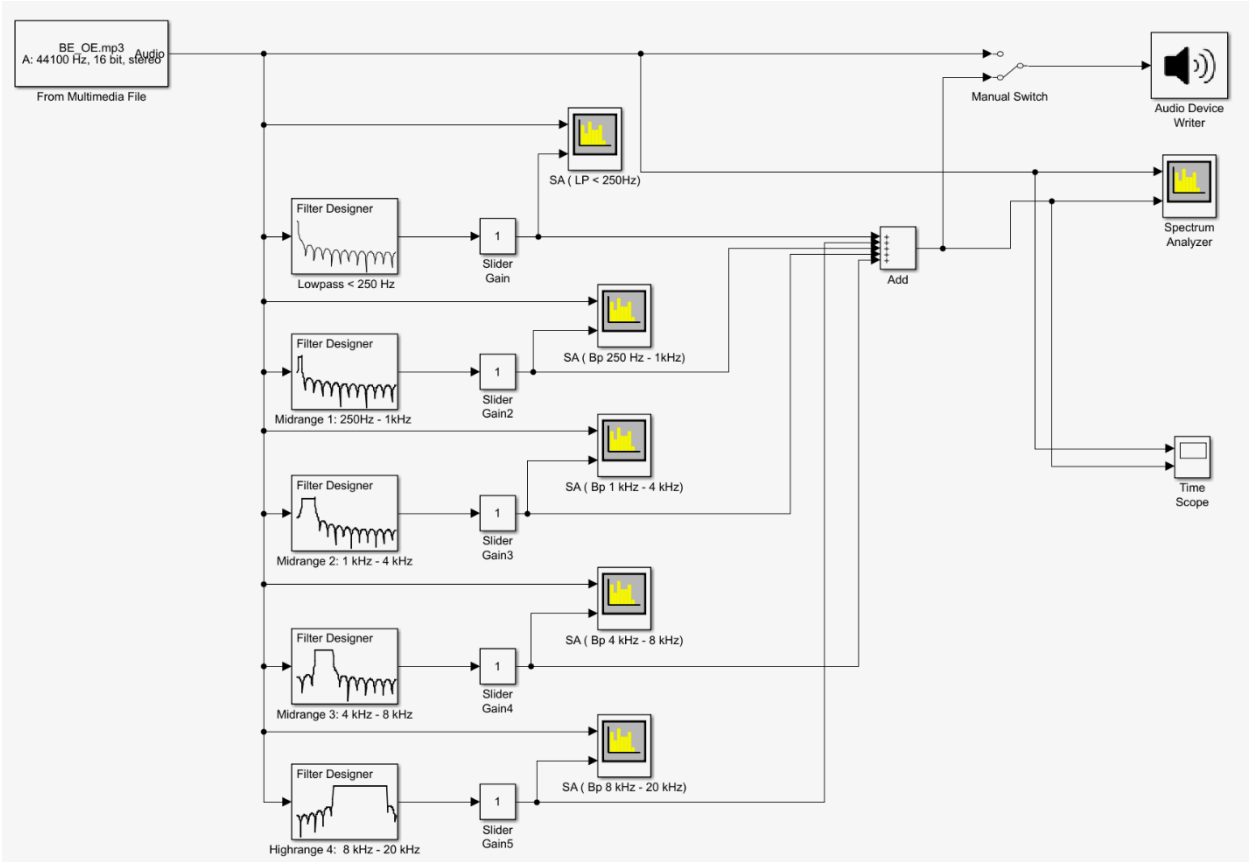
14th week

15th week
Final Exam



Time_Plan and Project (2/2)

1. Audio Equalizer using MATLAB
2. Audio-Noise Cancellation using MATLAB



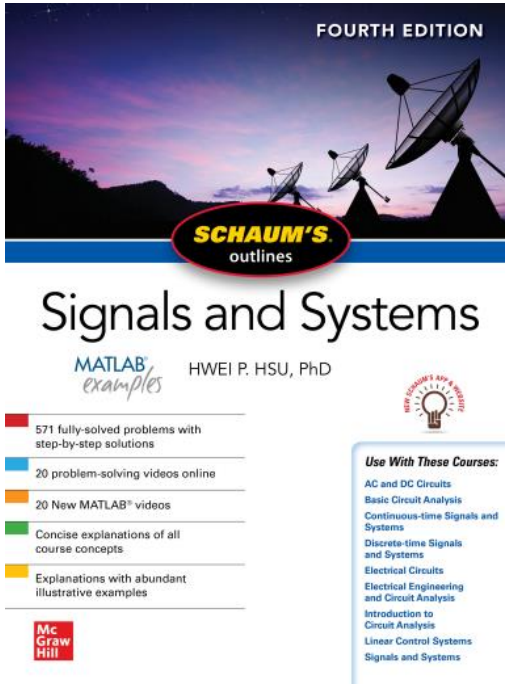
Useful Links: https://github.com/splAcharya/AudioEqualizerMatlab_Simulink

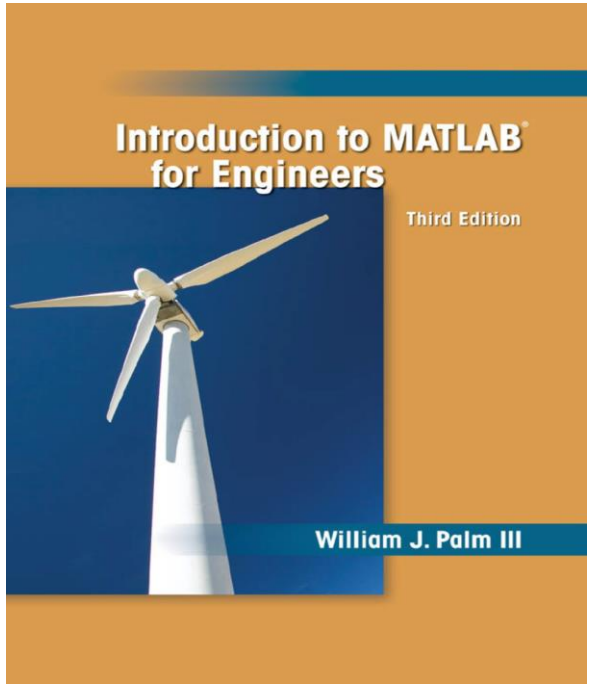
Outlines

- ❑ Course Information
- ❑ Course Assessment
- ❑ Course Content
- ❑ Course Long-Term-Vision
- ❑ IC Industry In Egypt
- ❑ References

References

- Hwei Hsu, ***Schaum's Outline of Signals and Systems***, *4th Ed.*, McGraw Hill, 2019.
- W. Palm "Introduction to MATLAB for Engineers," 3rd -McGraw-Hill Science_Engineering_Math, 2010.
- https://ece.engr.uvic.ca/~frodo/courses/ece260/#support_open_access
- http://www.ws.binghamton.edu/fowler/Fowler%20Personal%20Page/EECE301%20-%20Flipped.htm





Stay on target



[100%]

- ❑ Lectures
- ❑ Sections
- ❑ Labs
- ❑ Project

Thanks!